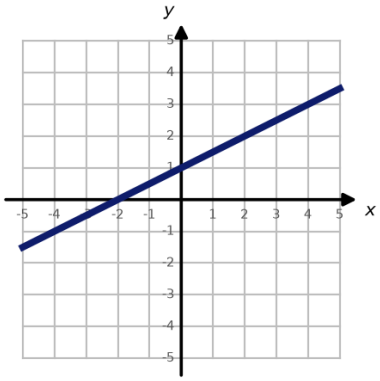
Linear Functions & Slope

Name: _____ Date: _____ Period: _____

Part 1 — Is each graph a linear function? Explain why or why not.

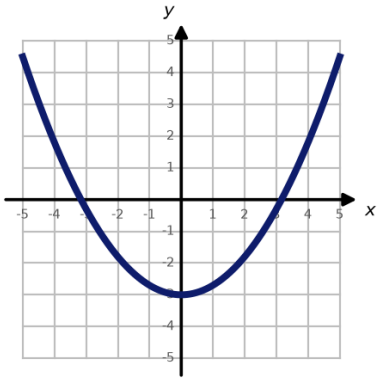
1. Is the graph a linear function?



Linear function? **Yes** **No**

Why? _____

2. Is the graph a linear function?



Linear function? **Yes** **No**

Why? _____

Part 2 — Is each table a linear function? Explain why or why not.

3. Is the table a linear function?

x	y
0	1
1	4
2	7
3	10
4	13

Linear function? **Yes** **No**

Why? _____

4. Is the table a linear function?

x	y
0	0
1	1
2	4
3	9
4	16

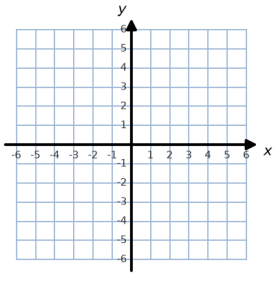
Linear function? **Yes** **No**

Why? _____

Part 3 — Complete each table, then plot the points and draw the line.

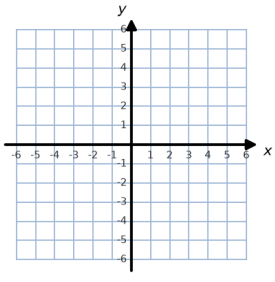
5) y = 2x - 1

X	Y	(x, y)
-2		
-1		
0		
1		
2		



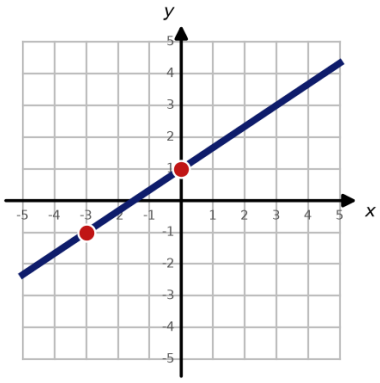
6) y = -x + 2

X	Y	(x, y)
-2		
-1		
0		
1		
2		



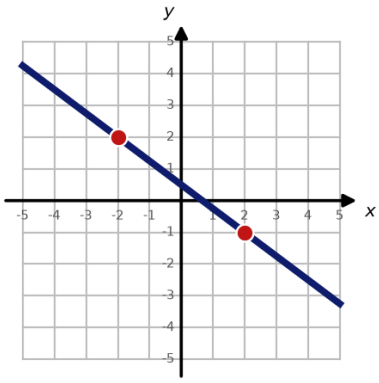
Part 4 — Find the slope of each line from the graph.

7. Find the slope of the line.



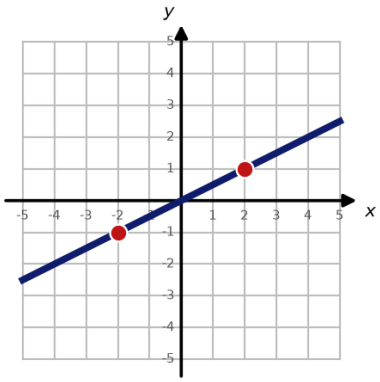
$m = \frac{\text{rise}}{\text{run}} = _____$

8. Find the slope of the line.



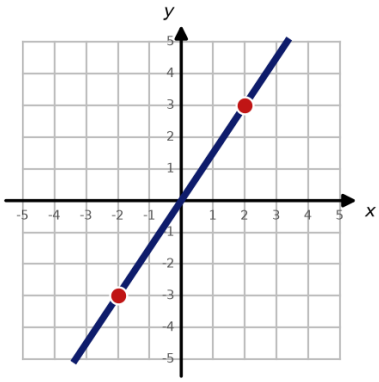
$m = \frac{\text{rise}}{\text{run}} = _____$

9. Find the slope of the line.



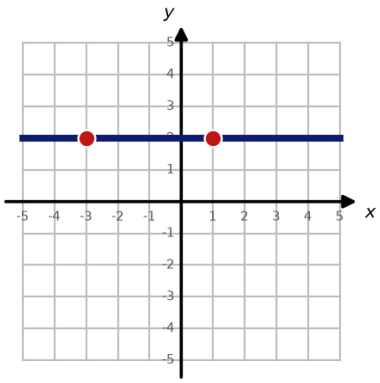
$m = \frac{\text{rise}}{\text{run}} = _____$

10. Find the slope of the line.



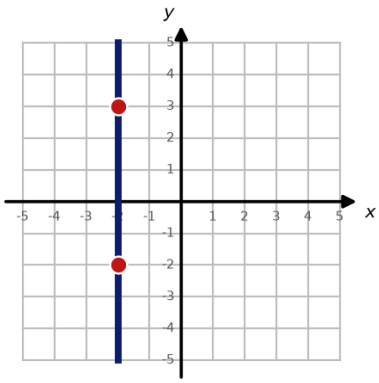
$m = \frac{\text{rise}}{\text{run}} = _____$

11. Find the slope of the line.



$m = \frac{\text{rise}}{\text{run}} = _____$

12. Find the slope of the line.



$m = \frac{\text{rise}}{\text{run}} = _____$

Part 5 — Find the slope between the two points. Simplify if needed.

13. Find the slope, simplify (reduce) if needed.

Given the points (1, 3) and (5, 11)

$$\frac{y_2 - y_1}{x_2 - x_1}$$

m = _____

14. Find the slope, simplify (reduce) if needed.

Given the points (-2, 1) and (3, -4)

$$\frac{y_2 - y_1}{x_2 - x_1}$$

m = _____

15. Find the slope, simplify (reduce) if needed.

Given the points (0, -3) and (4, 3)

$$\frac{y_2 - y_1}{x_2 - x_1}$$

m = _____

16. Find the slope, simplify (reduce) if needed.

Given the points (-4, 2) and (2, -1)

$$\frac{y_2 - y_1}{x_2 - x_1}$$

m = _____